

Autonomous Mobile Manipulator for Data Center Rack Stacking

Problem & Motivation

Data centers are expanding quickly, but server installation and replacement depend heavily on manual labor. **40%** of organizations suffered an outage caused by human error in the past 3 yrs, and **85%** are due to staff not following procedures [1].

What a successful run looks like:

Signal is sent to the husky (i.e Rack #3 shelf #2 unstack) -> husky autonomously navigates to rack -> image is scanned of the AprilTag (rack info and calibration) -> arm stack/unstacks -> Goes home

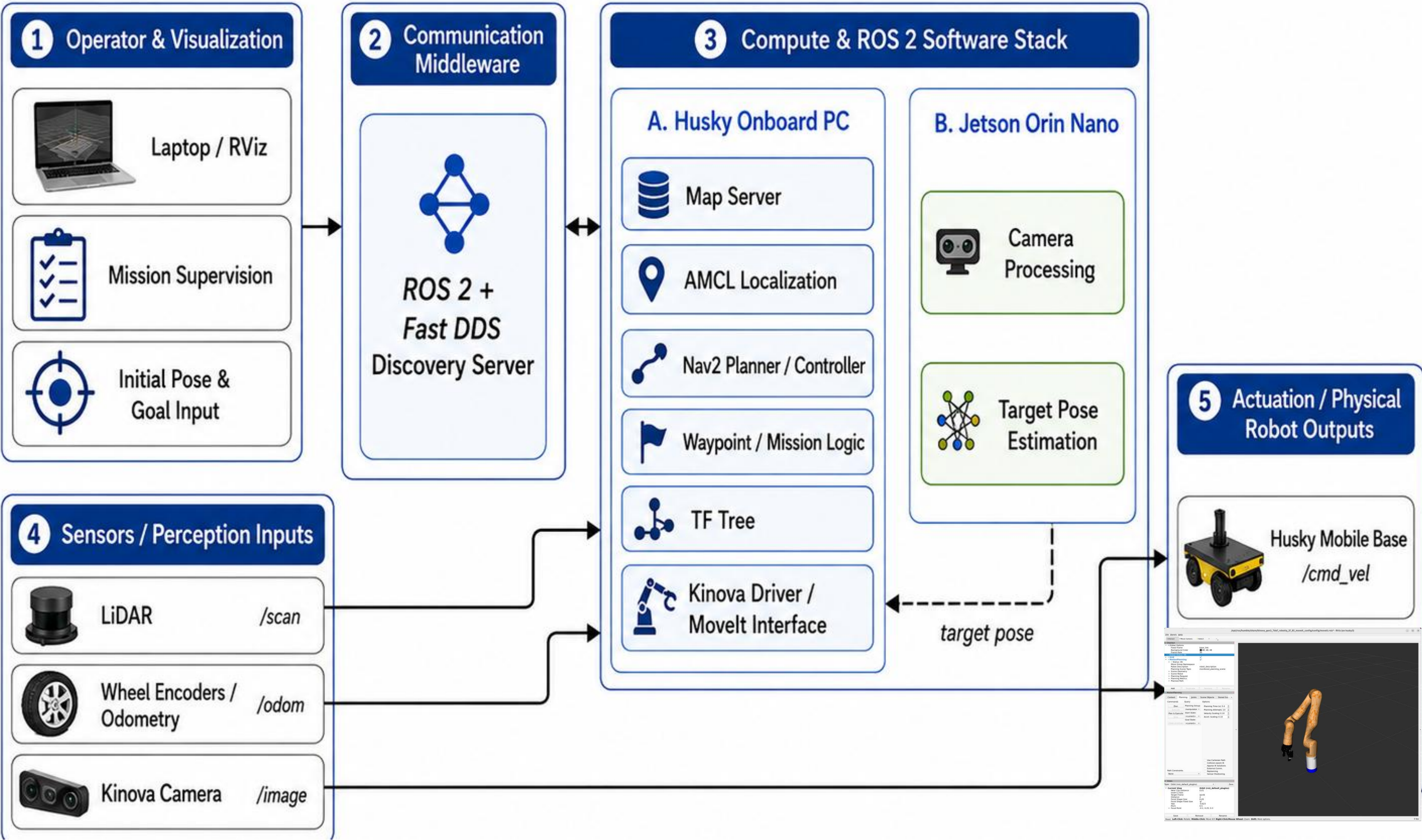
The Process from 0 to 100

Challenges solved:

- Network & connection
- Hardware wear/tear
- System integration
- Outdated software and documentation
- Camera Publisher connection



System Architecture



Applied Course Concepts

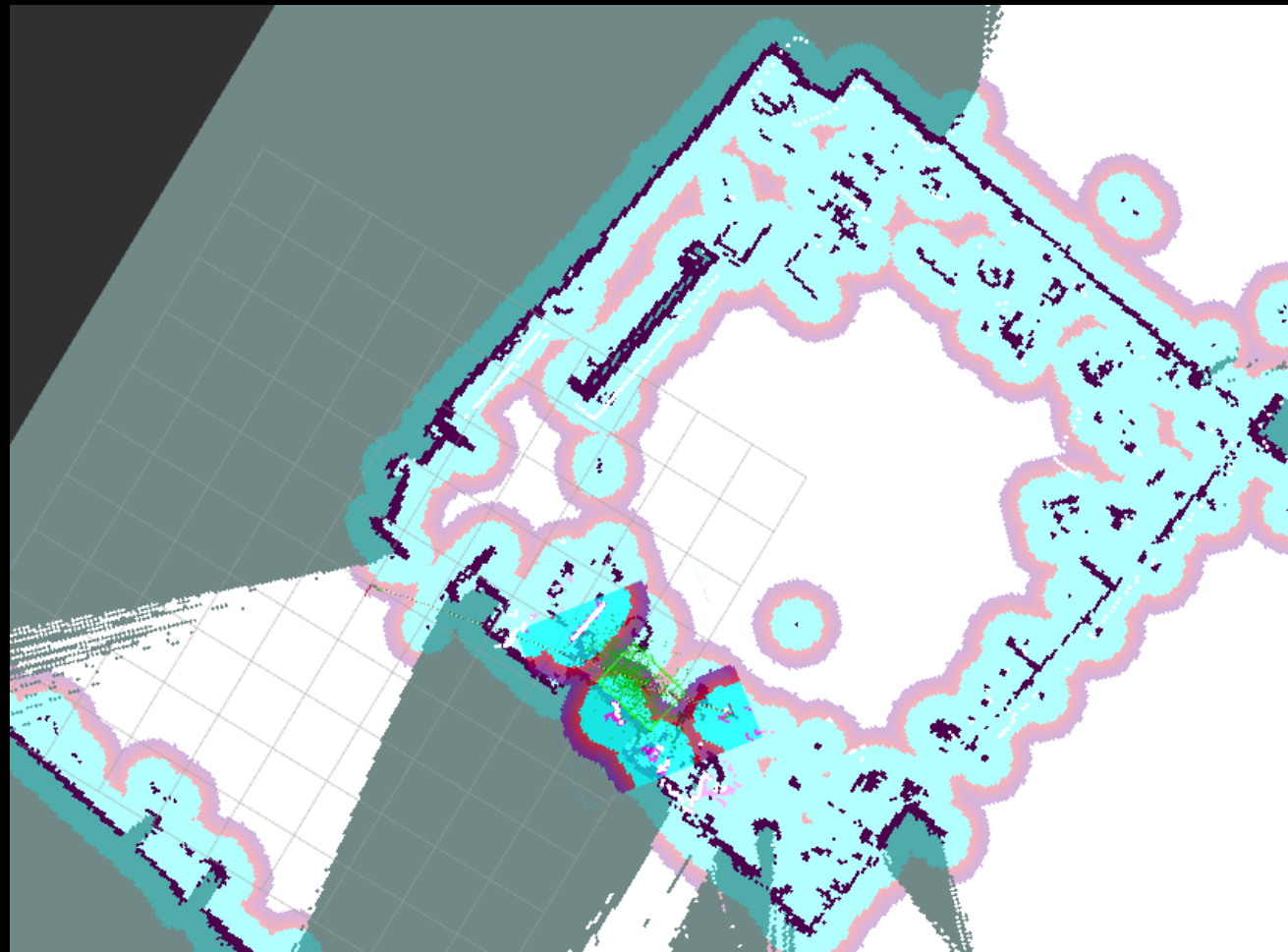
- Localization: AMCL is a particle-filter method matching LiDAR scans to map.
- Planning + control: separate global route planning from local trajectory tracking.
- Sensor fusion: odometry and joint states are proprioceptive; LiDAR and RGB camera are exteroceptive.
- Systems integration: lifecycle nodes, QoS and discovery.
- Frame transformations: Lidar, arm and camera to the base of the husky through ROS2.

Manipulation & Calibration

- AprilTag identifies the target rack and module ID.
- MoveIt 2 generates collision-free trajectories for the Kinova Gen3 manipulator.
- Predefined arm poses are manually calibrated to achieve repeatable grasping.
- Successful manipulation is verified by repeatable trajectory execution and stable grasping performance.

Navigation

- AMCL estimates $p(x,y,\theta)$ by comparing LiDAR scans to the map while odometry predicts motion.
- The global planner chooses the route; the local controller tracks it while costmaps block occupied cells.
- SLAM created a 2D occupancy-grid map of the racks, walls, and free space.



Evaluation & Results

Metric	Result
AprilTag Detection Rate	100% (Lab Tests)
Motion Planning Success	>95%
Navigation Success	>95%
ARM Start-up Time	~5–6 s
Grasp Success Rate	100%
Object Avoidance Success	85%

Table 1: Results for 15 trials run evaluating 6 metrics.

The system performed reliably in perception, motion planning, navigation, and grasping, but the lower 85% obstacle-avoidance success rate reveals the main weakness and safety risk because of the arm blocking it.

Progress

Weeks	Main Tasks	
1–2	Define the mock data-center rack plug-in scenario, confirm hardware availability, and set up ROS 2, robot model, coordinate frames, and basic Husky control.	✓
3–4	Set up sensors, including LiDAR and camera. Test RViz visualization, build or load a simple map, and prepare the mock rack and plug-in slot.	✓
5–6	Implement navigation to the target rack using Nav2. Add simple perception to detect the target rack or slot using a QR code, AprilTag, or colored marker.	✓
7–8	Set up Kinova arm motion planning and test low-speed plug-in motion on the mock rack. Integrate base alignment, target detection, and arm motion into one task sequence.	✓
9–10	Run experiments, collect evaluation data, and prepare the final demo, poster, presentation, and report.	✓ +

Table 2: Timeline and Milestone Progress.

Obstacle detection & Avoidance!



Future Work & Limitations

Future work could include improving on limitations:

- 5G or more stable WIFI to improve ROS 2 stability.
- Network latency reaches tens of ms, and the packet loss can be quite high as well.
- Lidar position and filter improvement.
- Navigation accuracy depends on map quality.
- Arm trajectories are not always optimal.
- Reliable hardware and updated software by Kionva and Clearpath.